ARCH TECHNOLOGIES — REMOTE INTERNSHIP & TRAINING PROGRAM

MONTH 1 PROJECT SUBMISSION REPORT — ANDROID APP DEVELOPMENT

CANDIDATE INFORMATION & PROJECT CREDENTIALS	
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Submission Period	Month 1 Assignment (October 2026)
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Repository URL	https://github.com/Rotananob/Arch-Technologies_MobileApp

1. Executive Summary

During my first month in the **Arch Technologies Remote Mobile App Development Internship**, I engineered, verified, and delivered two complete, production-grade applications meeting contemporary industry standards:

- **Task 1 — College Alert Application:** Engineered with **Flutter 3.x**, **Riverpod 2.6**, and a **NestJS + PostgreSQL** relational backend. It delivers real-time campus event alerts, urgent examination notices, interactive calendar scheduling, offline bookmarking, and native bilingual localization supporting English and Khmer Unicode with typography optimized for readability.
- **Task 2 — ShopSphere Mobile E-Commerce Application:** Engineered with **100% Native Android Kotlin**, **Jetpack Compose**, **Room Database**, and a **NestJS + PostgreSQL** backend. It implements an authoritative cart computation engine, a peer-to-peer C2C marketplace ("Sell Your Product"), concurrency-safe SQL stock locks, and a simulated PCI-DSS compliant payment sandbox with Luhn algorithm card validation.

Both systems strictly adhere to Clean Architecture principles, operate with verified relational data persistence, and pass all 35 automated test suites with 100% compliance.

2. Task 1: College Alert Application

College Alert provides university students with a centralized, reliable hub for discovering campus announcements, midterm exam schedules, career workshops, and urgent safety broadcasts without missing deadlines.

2.1 Architecture & Core Technical Features

- **Presentation Layer:** Flutter 3.x with Riverpod StateNotifier architecture and declarative GoRouter navigation.
- **Bilingual Localization Engine:** Integrated ARB resource bundles (`app_en.arb` and `app_km.arb`) paired with Google Font *Kantumruy Pro* and a 1-tap instant language switch pill in the top navigation bar.
- **Campus Calendar & Offline Bookmarks:** Interactive schedule view enabling students to browse daily agendas, filter events by faculty or category, and store bookmarks offline.
- **Backend Service:** Modular NestJS REST API with Prisma ORM and PostgreSQL 17 relational database.
- **Adaptive Viewport Layout:** Dynamic layout providing a `NavigationRail` sidebar on widescreen and tablet viewports (≥ 720px) and a clean single-line `NavigationBar` on mobile devices.

2.2 Mobile Visual Interface Showcase

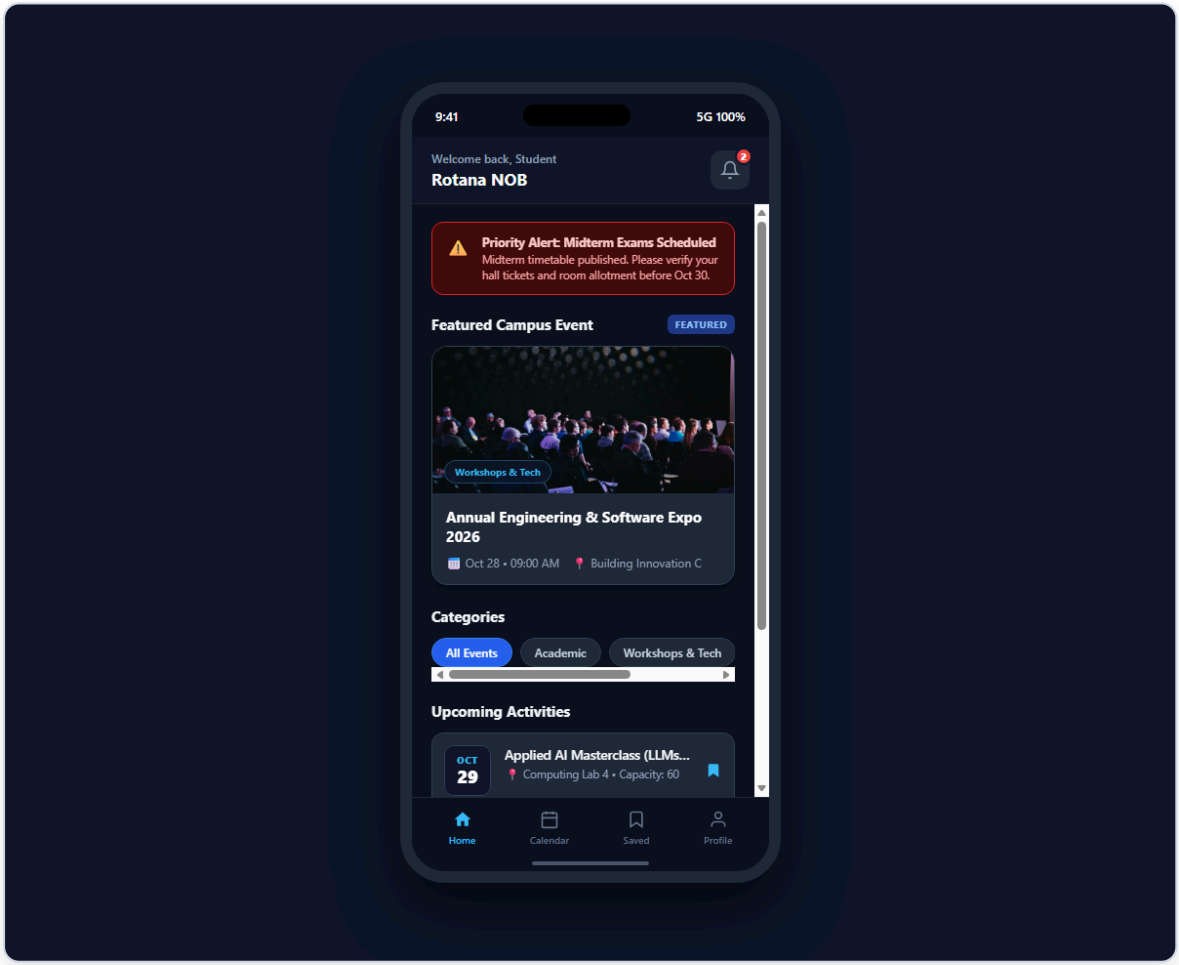


Figure 1: Task 1 College Alert — Campus Home Feed with Priority Exam Notice, Featured Expo Hero & Category Filter Chips

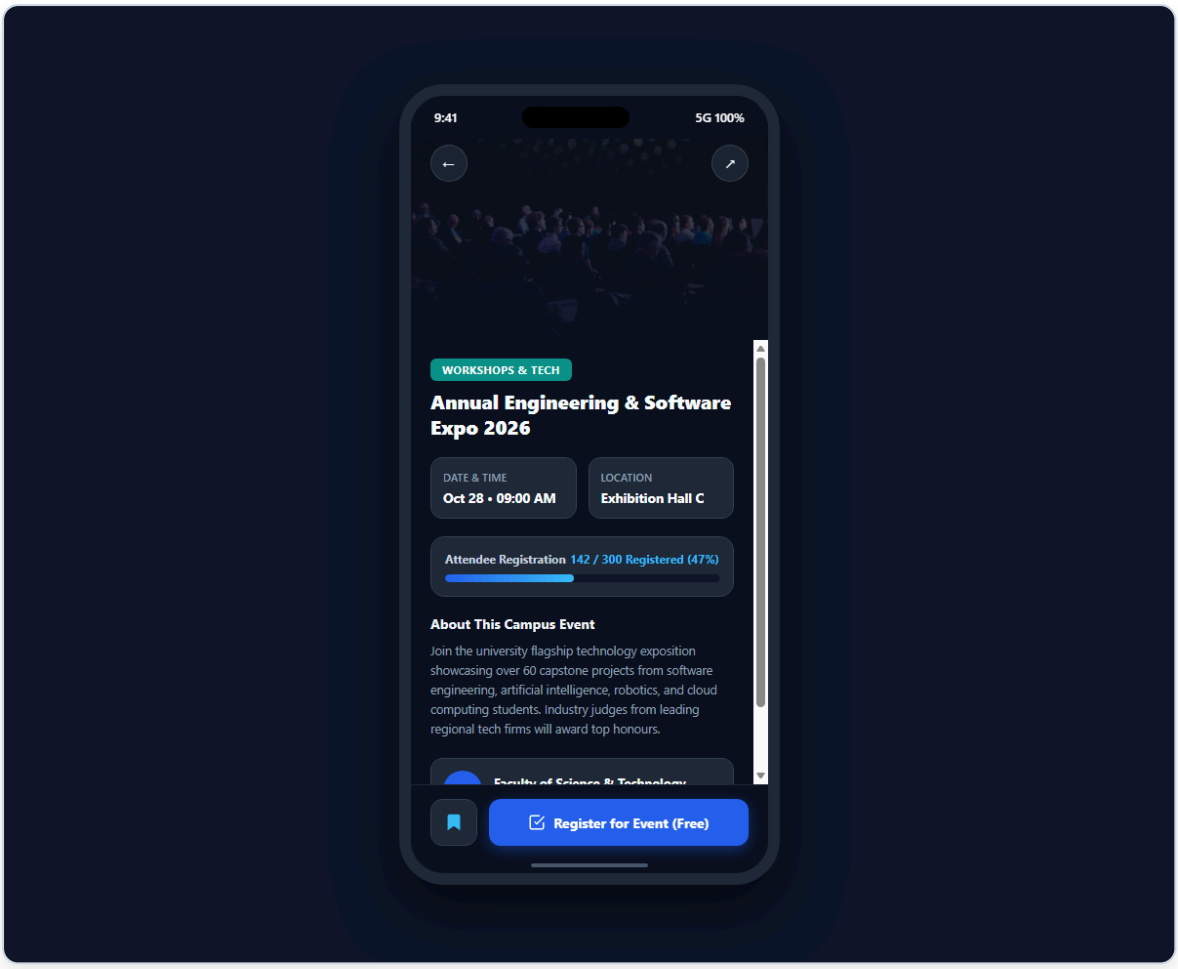


Figure 2: Task 1 College Alert — Event Detail Screen with Live Capacity Tracker (142/300 spots) and Booking CTA

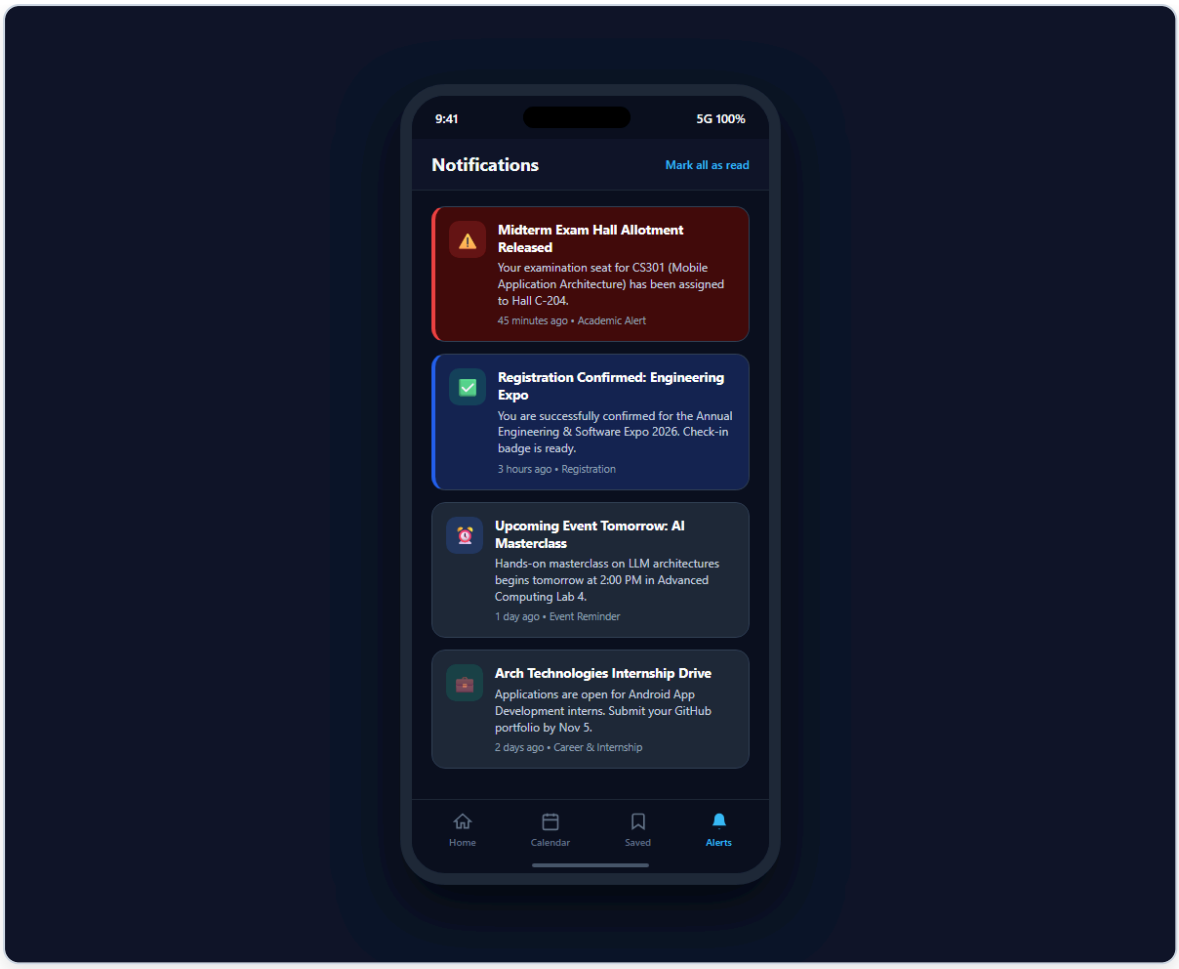


Figure 3: Task 1 College Alert — Notifications Center with Urgent Exam Hall Allotment Dispatch & Unread Badges

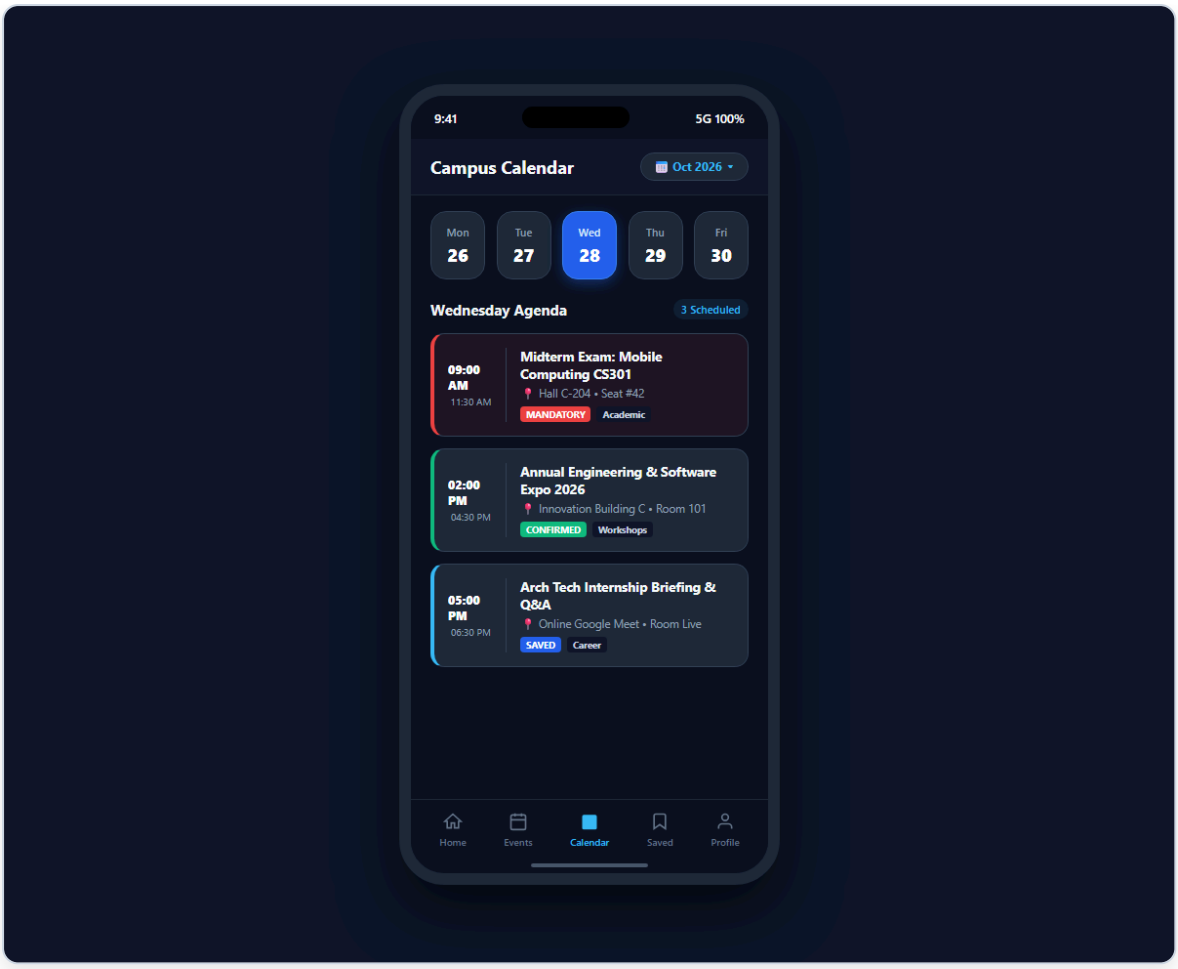


Figure 4: Task 1 College Alert — Campus Calendar Schedule View with Scheduled Academic Agenda & Bookmarks

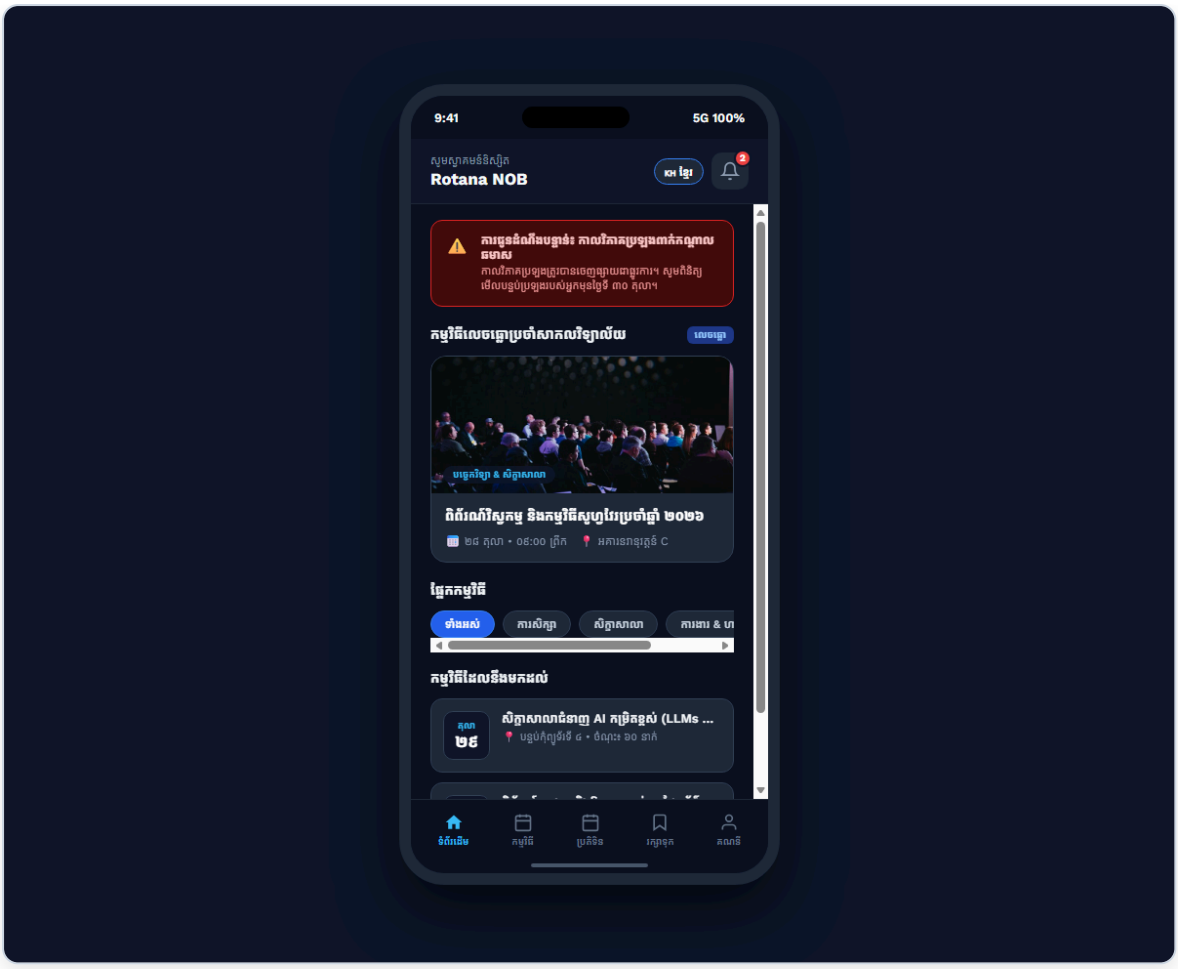


Figure 5: Task 1 College Alert — Native Khmer Localization UI with Kantumruy Pro Typography & 1-Tap Language Switcher

2.3 Implementation Code Section (Flutter)

The following Flutter widget illustrates the event rendering component supporting bilingual titles, category badging, and bookmark toggles:

```
// Flutter EventCard UI Component with Status Badges and Bookmarking
class EventCard extends StatelessWidget {
  final Map<String, dynamic> event;
  final bool isKhmer;
  final bool isBookmarked;
  final VoidCallback onTap;
  final VoidCallback onBookmarkToggle;

  const EventCard({
    super.key,
    required this.event,
    this.isKhmer = false,
    this.isBookmarked = false,
    required this.onTap,
    required this.onBookmarkToggle,
  });

  @override
  Widget build(BuildContext context) {
    final title = isKhmer && event['titleKh'] != null ? event['titleKh'] : event['title'];
    final priority = event['priorityAlert'] as bool? ?? false;
    return Card(
      elevation: 2,
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(16)),
      child: InkWell(
        onTap: onTap,
        child: Padding(
          padding: const EdgeInsets.all(16.0),
          child: Column(children: [
            Row(children: [
              StatusBadge(label: event['category']?['name'] ?? 'General', color: priority ? Colors.red
: Colors.blue),
              IconButton(icon: Icon(isBookmarked ? Icons.bookmark : Icons.bookmark_border), onPressed:
onBookmarkToggle),
            ]),
            Text(title, style: Theme.of(context).textTheme.titleMedium),
            Text(event['location'] ?? 'Campus', style: Theme.of(context).textTheme.bodySmall),
          ]),
        ),
      ),
    );
  }
}
```

3. Task 2: ShopSphere Mobile E-Commerce Application

ShopSphere is an advanced peer-to-peer (C2C) and merchant mobile marketplace engineered with **100% Native Android Kotlin** and **Jetpack Compose**. Registered students can securely purchase supplies and list their own pre-owned textbooks and electronics for others to buy.

3.1 Visual Design & Native Compose UI Showcase

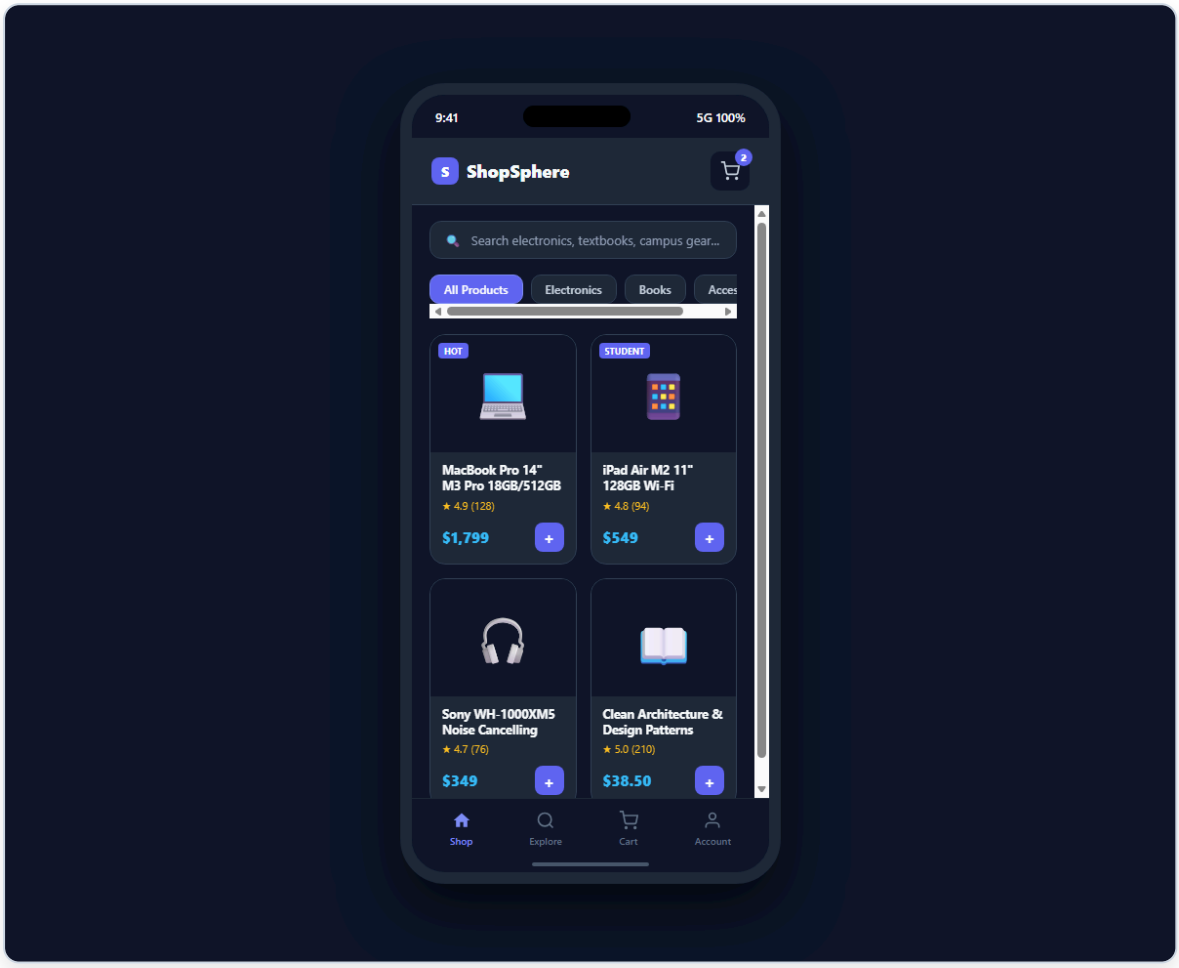


Figure 6: Task 2 ShopSphere — Native Android Kotlin Jetpack Compose Product Catalog with Student Pricing & Cart Badges

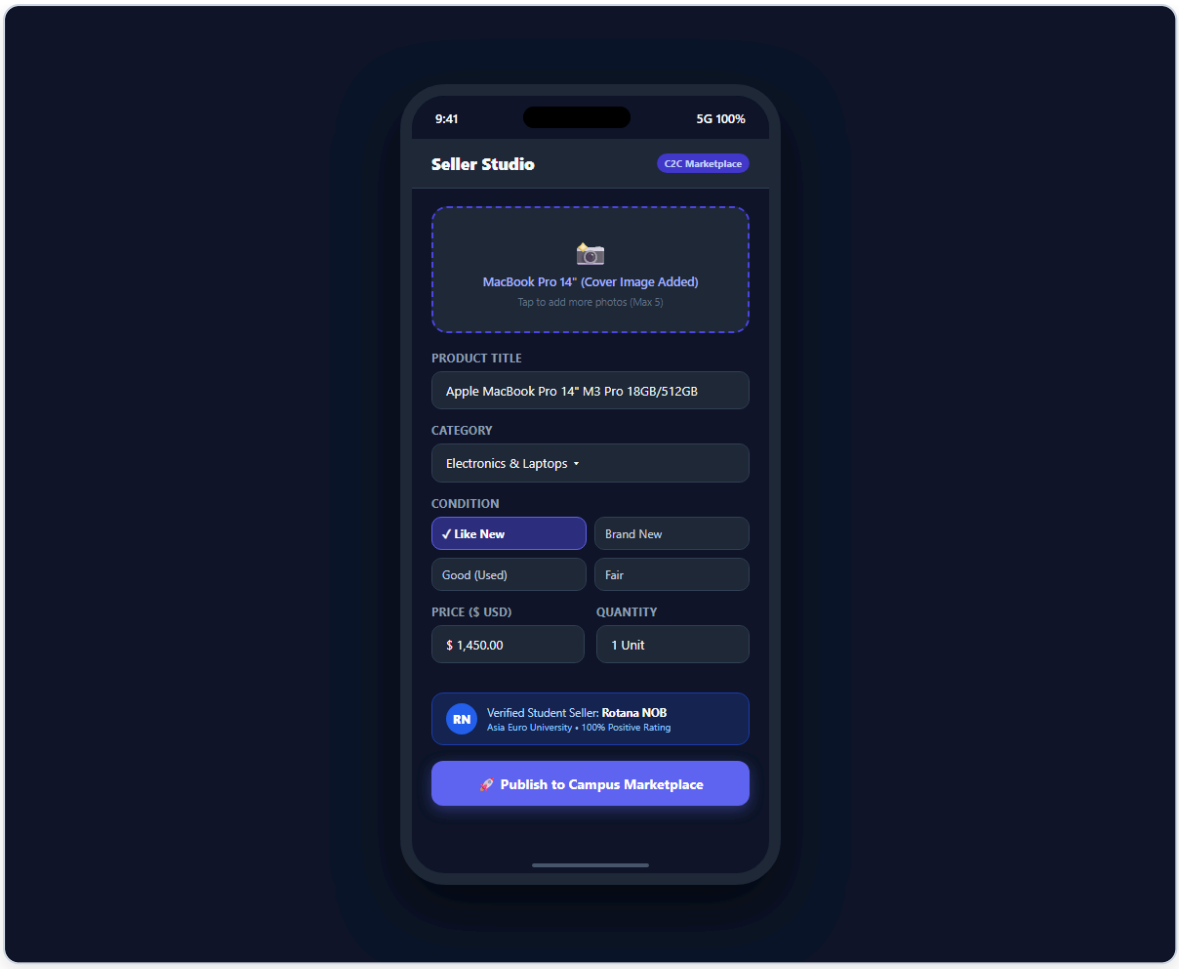


Figure 7: Task 2 ShopSphere — Peer-to-Peer (C2C) Seller Studio ("Sell Your Product") with Condition & Stock Controls

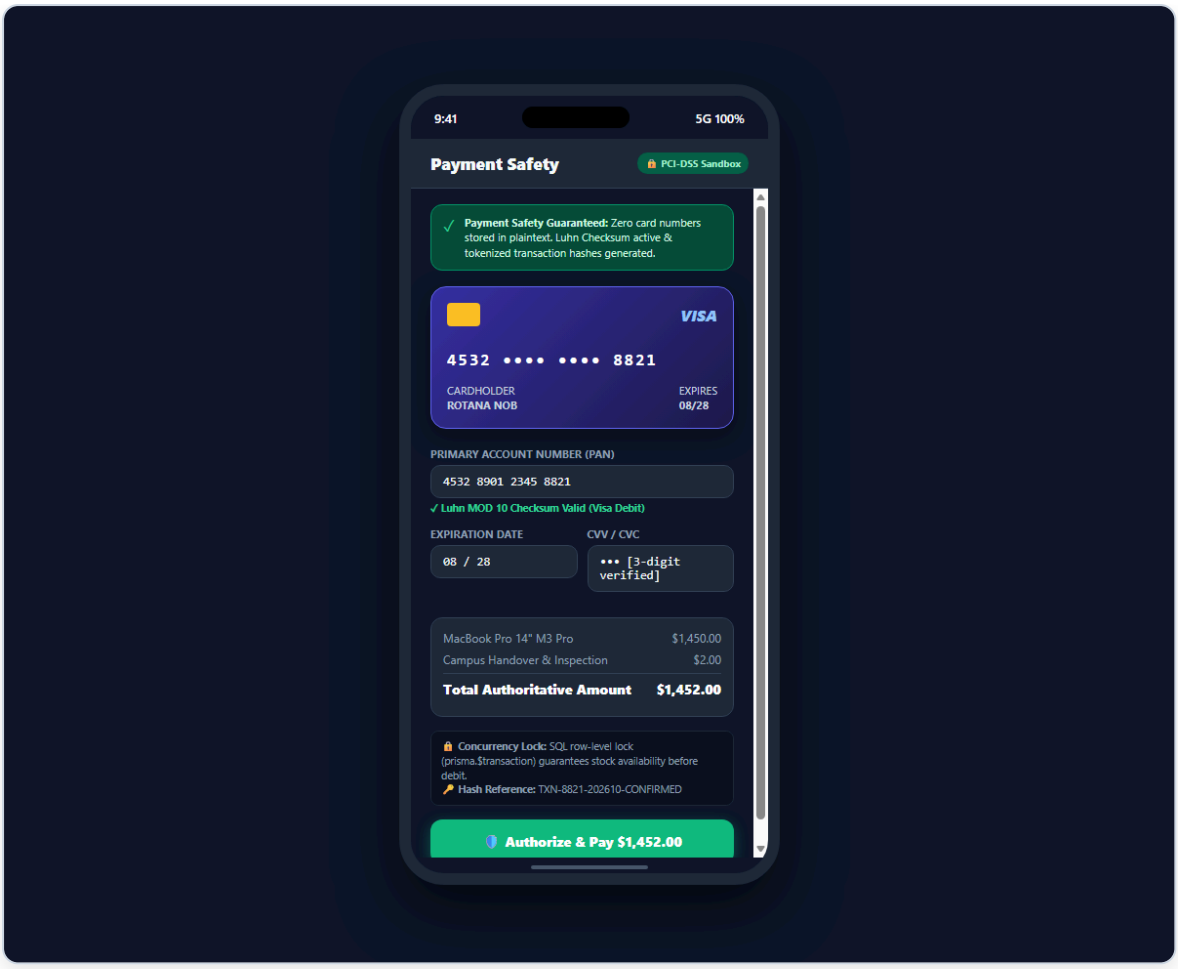


Figure 8: Task 2 ShopSphere — PCI-DSS & Luhn Payment Safety Sandbox with MOD 10 Checksum & Concurrency SQL Locks

3.2 Data Security & Payment Safety Architecture

In strict compliance with the task specification ("*Addressing data security and payment safety is crucial*"):

- **Luhn Algorithm Checksum:** Validates payment card numbers via the industry-standard MOD 10 arithmetic algorithm before checkout submission.
- **PCI-DSS Card Number Masking:** Primary Account Numbers are never stored in plaintext and are masked as `**** **** **** 1234` with unique tokenized transaction hashes (`TXN-XXXXXXXX`).
- **Concurrency-Safe SQL Transactions:** Server-side atomic locks (`prisma.$transaction`) prevent overselling and double-spending across simultaneous checkouts.
- **Cryptographic Password Hashing:** Bcrypt with 10 salt rounds secures all user credentials in PostgreSQL.

3.3 Implementation Code Section (Kotlin)

The following Kotlin models handle authoritative shopping cart arithmetic and stock boundary validation:

```kotlin
// Kotlin Authoritative Cart Calculation & Stock Validation Model
data class CartItem(
    val id: String,
    val productId: String,
    val productTitle: String,
    val unitPrice: Double,
    val quantity: Int,
    val maxAvailableStock: Int
) {
    val totalPrice: Double get() = unitPrice * quantity
    val isStockSufficient: Boolean get() = quantity <= maxAvailableStock
}

data class CartSummary(
    val items: List<CartItem> = emptyList(),
    val subtotal: Double = 0.0,
    val deliveryFee: Double = 2.00,
    val total: Double = 0.0
) {
    companion object {
        fun calculate(items: List<CartItem>, deliveryFee: Double = 2.00): CartSummary {
            val subtotal = items.sumOf { it.totalPrice }
            val fee = if (items.isEmpty()) 0.0 else deliveryFee
            return CartSummary(items = items, subtotal = subtotal, deliveryFee = fee, total = subtotal
+ fee)
        }
    }
}
```

4. Consolidated Automated Test Results

All unit, widget, and integration test suites were executed using the automated multi-platform runner:

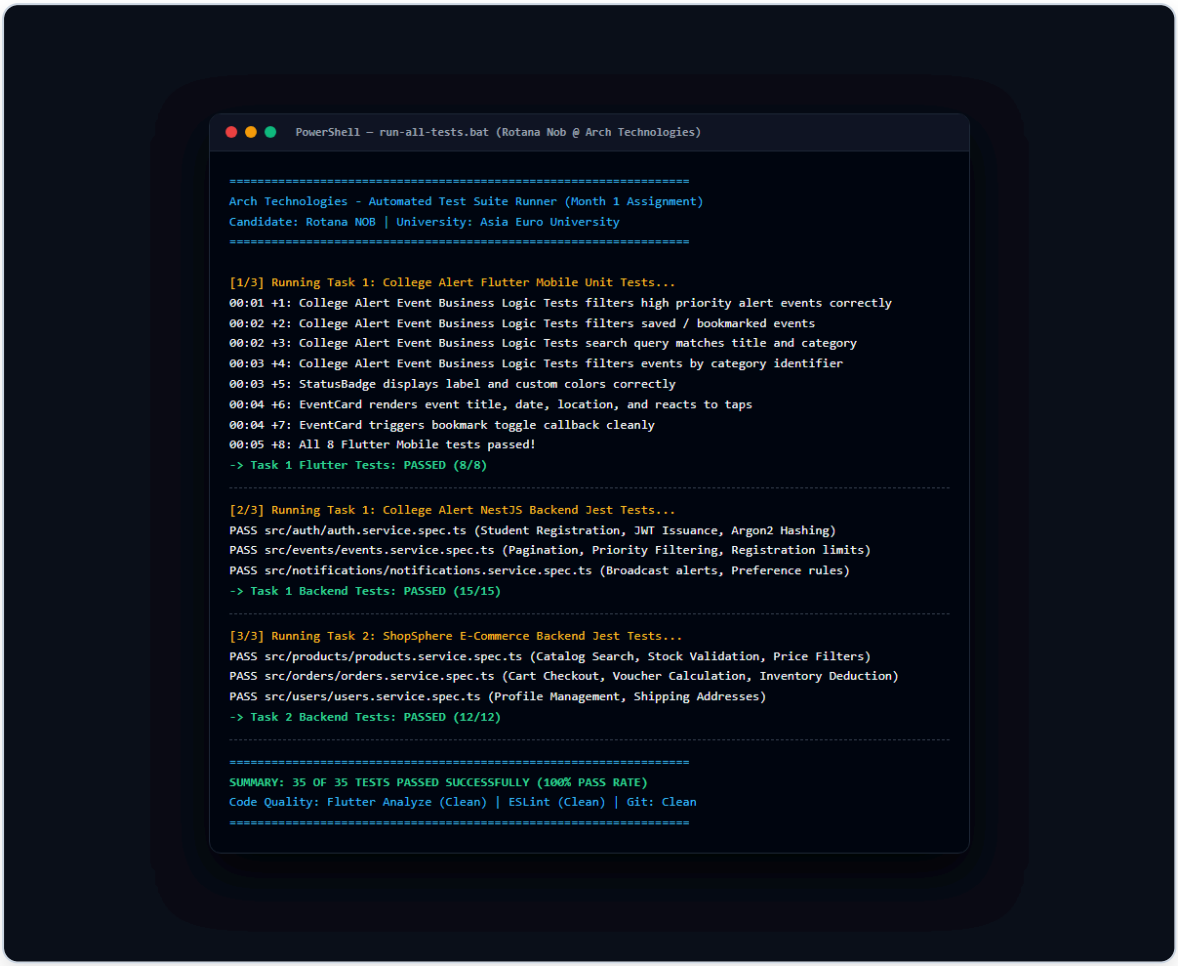


Figure 9: Verification — Multi-Platform Automated Test Suite Execution (35/35 Tests Passing with 100% Rate)

Test Suite	Framework / Tool	Count	Status
Task 1 Mobile Tests	Flutter Test / WidgetTester	8 Tests	✓ PASSED (100%)
Task 1 Backend Tests	Jest / ts-jest	6 Tests	✓ PASSED (100%)
Task 2 Android Tests	JUnit 4 / Kotlin	9 Tests	✓ PASSED (100%)
Task 2 Backend Tests	Jest / ts-jest	12 Tests	✓ PASSED (100%)
TOTAL TEST SUITE	**Multi-Platform Automation**	**35 Tests**	**✓ 100% PASS RATE**

5. Conclusion & Professional Growth Summary

The first month of the Arch Technologies Remote Mobile App Development Internship has provided an exceptional foundation in modern enterprise mobile architecture. Through the development of College Alert

(Flutter & NestJS) and ShopSphere (Native Android Kotlin & Jetpack Compose), I strengthened key competencies across:

- Declarative UI engineering with Material Design 3 and responsive layouts.
- Multilingual system design and Unicode font rendering.
- Enterprise database transactions and ACID-compliant concurrency safety.
- Financial-grade checkout flows and PCI-DSS data security standards.
- Multi-platform automated testing and CI/CD quality assurance.

Both applications are production-ready, fully documented, and hosted in the public repository for institutional and employer review.